

Academic Statement of Purpose

Program: Master of Science in Robotics

Applicant: Yang Sen (Liam) Lin

*I aspire to architect “**Trustworthy Embodied Systems**” for aerial robotics by mathematically unifying the adaptability of Vision-Language Models with the provable safety guarantees of non-linear control.*

In my recent research accepted to **CoRL 2025**, titled “**See, Point, Fly**,” I identified a fundamental tension in the current state of robotic autonomy: geometric control provides reliability but becomes brittle in unstructured environments; conversely, Generative Vision-Language-Action (VLA) models offer adaptability but suffer from hallucinations and high computational latency. My objective at the University of Michigan is to resolve this tension. I aim to investigate the mathematical interfaces between rigid control theory and probabilistic foundation models, leveraging the Robotics Institute’s unique “**Sensing, Reasoning, Acting**” paradigm. Specifically, I seek to bridge the embodied language grounding research of **Prof. Joyce Chai** with the safety-critical control frameworks of **Prof. Dimitra Panagou** to build aerial agents that are both semantically intelligent and mathematically safe.

My journey toward this objective began with a focus on algorithmic decoupling. In my **CoRL 2025** work, I formulated a geometric projection algorithm to ground 2D VLM prompts into 3D metric controls, effectively solving depth ambiguity via pin-hole camera models without external sensors. By framing action prediction as a spatial grounding task rather than end-to-end text generation, I achieved **state-of-the-art success rates (93.9%)** in unstructured environments. However, through rigorous testing, I observed that my “learning-free” geometric priors became brittle when facing erratic, dynamic obstacles where non-linear reasoning is required. To overcome this “**reactivity bottleneck**,” I realized I must evolve from static heuristics to embodied generalist policies. This realization drives my application to UMich, where I specifically seek to master the intersection of **Reasoning (AI)** and **Acting (Control)**.

Simultaneously, my research philosophy aligns with UMich’s “Full-Stack” roboticist mindset—I believe algorithmic innovation must be grounded in physical reality. While many researchers focus solely on high-level inference, my internship at **Wolley Inc.** grounded me in the constraints of hardware architecture. There, I engineered C-based diagnostic tools to profile **CXL memory bandwidth** and host pressure. This experience gave me the low-level architectural intuition needed to optimize I/O efficiency for physical AI. I plan to extend this hardware-software co-design mindset during my upcoming internship at **OMRON SINIC X**, where I will distill Robotics Foundation Models for edge deployment. I recognized early on that distinct algorithmic layers are insufficient without optimizing the entire system—from memory protocols to control loops. This makes UMich, with its emphasis on integrated systems, the ideal academic home for my work.

My academic preparation is defined by the convergence of **Mechanical Engineering and Computer Science**, mirroring the interdisciplinary nature of the Robotics Institute. While ME grounded me in the physical first principles of system dynamics, my dual focus in CS equipped me with the algorithmic reasoning of modern AI. I demonstrated the ability to synthesize these fields at a graduate level by securing an **A+ in Graduate Computer Vision**. Furthermore, I have actively applied advanced mathematics to solve engineering problems, utilizing Projective Geometry to derive non-linear unprojection laws in my CoRL paper.

However, I recognize that defining safety for probabilistic manifolds requires tools beyond deterministic calculus. I am eager to challenge myself with **ROB 501: Mathematics for Robotics** to master Convex Optimization and Lie Group Theory. I view ROB 501 not just as a requirement, but as the essential theoretical bootcamp that will enable me to formalize rigorous safety constraints for next-generation aerial systems. Additionally, to further hone my engineering capabilities, I plan to take **ROB 550: Robotic Systems Laboratory**, ensuring my theoretical models can withstand the realities of physical deployment.

My technical roadmap directly maps to the research being conducted at the **Ford Robotics Building**. To address the “Reasoning” aspect of my dual objective, I aim to work with **Prof. Joyce Chai** in the **SLED Lab**. I am particularly interested in her work on embodied language grounding. I intend to take her course, **EECS 692: Advanced Artificial Intelligence**, to research how to ground high-level semantic instructions into actionable physical primitives, directly addressing the “VLM hallucination” issues I encountered in my previous work.

Simultaneously, to address the “Acting” aspect, I seek to collaborate with **Prof. Dimitra Panagou** in the **DASC Lab**. Her expertise in safety-critical control is the missing piece in my current research. By enrolling in **EECS 562: Nonlinear Systems and Control**, I hope to learn how to design **Control Barrier Functions (CBFs)** that act as “safety filters” for VLM outputs. My specific goal is to address the frequency mismatch in embodied systems: synthesizing a multi-rate control architecture that enforces high-frequency safety constraints (500Hz) despite the inference latency of large foundation models (0.5Hz).

Beyond the laboratory, I resonate deeply with the department’s motto, “**Robotics for the People.**” As the **GDSC Chapter Lead**, I worked to democratize access to technical education, organizing free workshops that impacted over 40 students from underrepresented and non-CS backgrounds. I aimed to dismantle the “elitist” barrier often associated with AI development. I aspire to bring this spirit of inclusive innovation to UMich, fostering a collaborative culture within the Ford Robotics Building where “serving society” is the metric of success, not just citation counts.

Ultimately, the UMich MS in Robotics will transform me from a robotics engineer who builds working prototypes into a **Computational Systems Architect** capable of defining the safety guarantees required to deploy autonomous agents. I am ready to contribute to the Michigan Robotics community, pushing the boundaries of what reliable, embodied AI can achieve for society.